

COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No. 06

**BOOLEAN THEOREMS AND
GATES**



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Recap of Previous Lecture

XOR & NAND gate





Topics to be Covered

NAND & NOR gates



H.W.

find out minimum no. 2-i/p NAND gate require to implement functions given below:

Q.1 $f(A, B, C) = \bar{A}B + A\bar{C} \longrightarrow \text{ANS: 4}$

Q.2 $f(A, B, C) = \bar{A}\bar{B} + A\bar{C} + B\bar{C} \longrightarrow \text{ANS: 4}$

Q.3 $f(A, B, C) = AB + BC + AC \longrightarrow \text{ANS: 6}$

Q.4 $f(A, B, C) = \overline{AB} + \overline{BC} \longrightarrow \text{ANS: 3}$

Q.5 $f(A, B, C, D) = \overline{AB} + \overline{CD} \longrightarrow \text{ANS: 5}$

Q.6 $f(A, B, C) = \overline{\bar{A}BC} \longrightarrow \text{ANS: 4}$

$$(\bar{A} + \bar{C})(A + B)$$

$$\bar{A}\bar{C}(A+B)$$

$$= R(A+B) = RA + RB$$

(IN)

$$\bar{f} = \overline{R_A} \quad \overline{R_B}$$

IN IN

$$f = P \cdot Q$$

$$f = \overline{P \cdot Q} \quad \text{IN}$$

$$Q.3 \quad f(A, B, C) = AB + BC + AC$$

$$\overline{f} = \overline{AB} \cdot \overline{BC} \cdot \overline{AC}$$

$$\begin{array}{ccc} \text{IN} & \text{IN} & \text{IN} \\ P & Q & R \end{array}$$

$$\overline{f} = PQR$$

$$f = \overline{PQR} \longrightarrow 3N \text{ (2-4P NAND)}$$

$$Q.4 \quad f(A, B, C) = \overline{AB} + \overline{BC} \quad \underline{3}$$

$$\overline{f} = AB \cdot BC = ABC$$

$$f = \overline{ABC} \longrightarrow 3 \text{ (2-4P)}$$

$$Q.5 \quad f(A, B, C, D) = \overline{AB} + \overline{CD} \quad \underline{5}$$

$$\overline{f} = AB \cdot CD$$

$$f = \overline{ABCD} = \overline{P \cdot Q} \text{ (1N)}$$

$$A \cdot B \longrightarrow 2N \rightarrow P$$

$$C \cdot D \longrightarrow 2N \rightarrow Q$$

Q. $f(A, B) = \bar{A}B + A\bar{B}$

where complement $\bar{A}$ & $\bar{B}$ available, then minimum no. of 2-i/p NAND gate requires to implement f 3.

$$f(A, B) = \bar{A}B + A\bar{B}$$

$$\bar{f} = \overline{\bar{A}B + A\bar{B}}$$

$$\bar{f} = \overline{\bar{A}B} \cdot \overline{A\bar{B}}$$

$$\bar{f} = \overline{\bar{A}B} \cdot \overline{A\bar{B}} \quad (1N)$$

[**Question**]

$$y = A\bar{B} + BC$$

Minimum no. of 2-input NAND gates required to implement y 4

Question

$$f(A, B, C) = \bar{A} + \bar{A}B + \bar{A}B\bar{C} = \bar{A} [1 + B + B\bar{C}] = \bar{A} \cdot 1 = \bar{A}$$

Minimum no. of 2-input NAND gates required to implement y 1

[Question]



$$y = \bar{A}BC$$

To implement y , minimum no. of 2-input NAND gates required is 5

$$y = \bar{A}BC$$

$$B \cdot C \longrightarrow 2N \longrightarrow P$$

$$y = \bar{A} \cdot P = Q \cdot P \quad (2N)$$

↓
Q (1N)

[Question]

$$f(A, B, C, D) = (\bar{A} + \bar{B})(C + D)$$

Minimum no. of 2-input NAND gates required to implement

$$f(A, B, C, D) \underline{\hspace{1cm} 4 \hspace{1cm}}$$

$$= \overline{\overbrace{AB}^{\substack{R \\ (1N)}}} (C+D) = RC + RD$$

$$\overline{f} = \overline{RC} \overline{RD}$$

(1N)
(1N)

P
Q

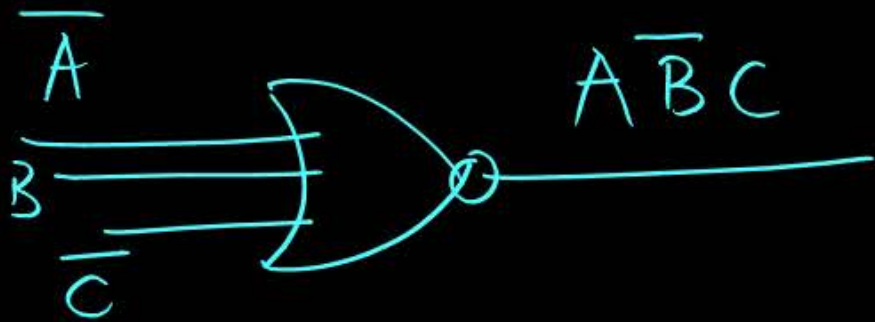
$$\overline{f} = P \cdot Q$$

$$f = \overline{P \cdot Q} \text{ (1N)}$$

[**NOR GATE**] OR+NOT



Representation :

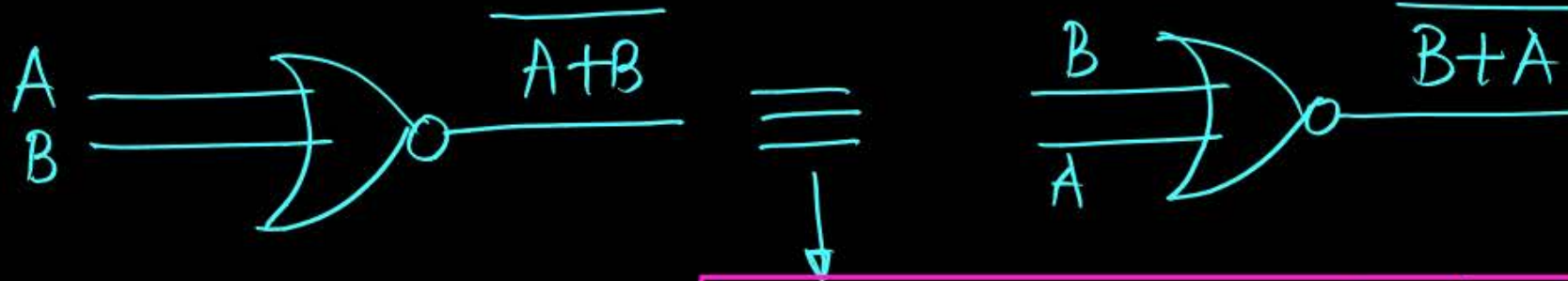


	A	B	$y = \overline{A+B}$
0	0	0	1
1	0	1	0
2	1	0	0
3	1	1	0

$$\begin{aligned}
 y(A, B) &= \sum (0) \\
 \overline{A} \cdot \overline{B} &= \pi(1, 2, 3) \\
 &= (A + \overline{B})(\overline{A} + B) \\
 &\quad (\overline{A} + \overline{B}) \\
 &= \overline{A}(A + \overline{B}) \\
 &= 0 + \overline{A} \cdot \overline{B} \\
 &= \overline{A} \cdot \overline{B}
 \end{aligned}$$



- Commutative Law :

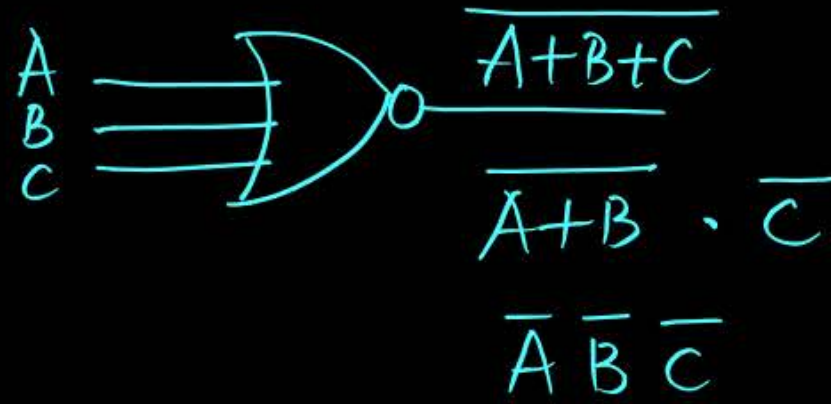


It holds commutative law

position of variables is irrelevant.

$$\overline{A+B} = \overline{B+A}$$

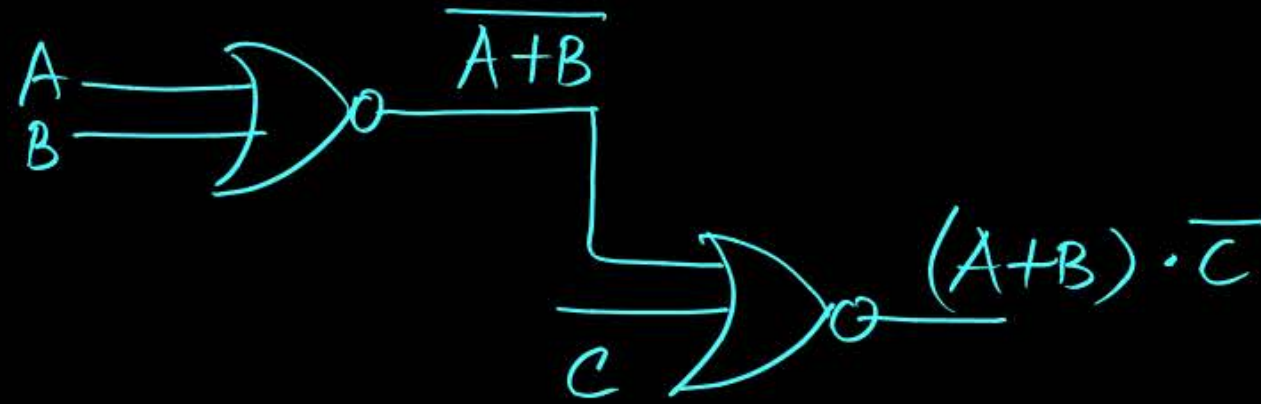
- Associative Law :



$\neq$

↓

It does not hold associative law.



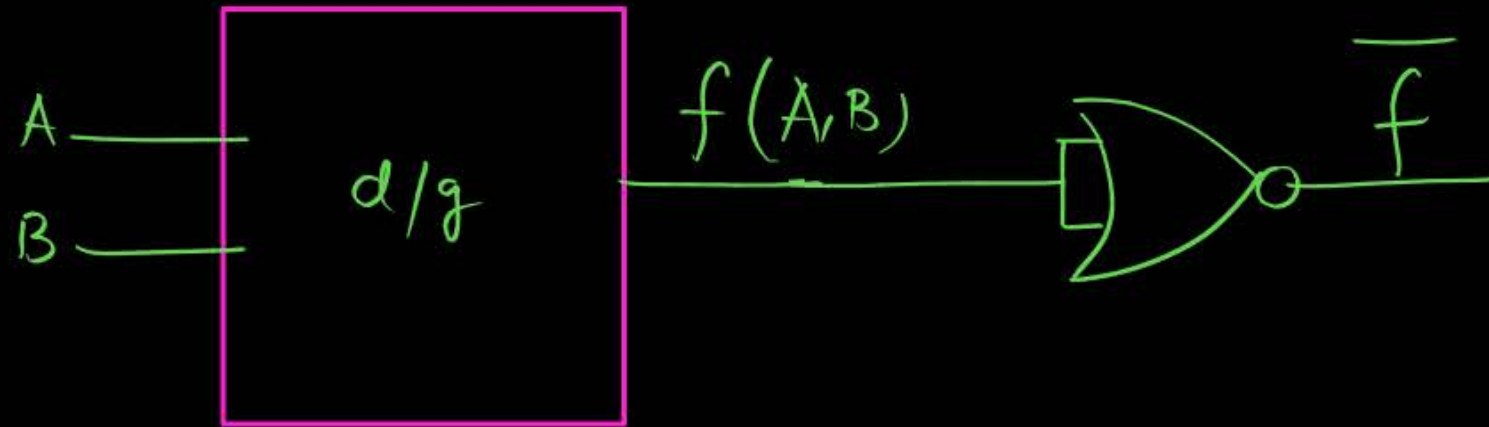
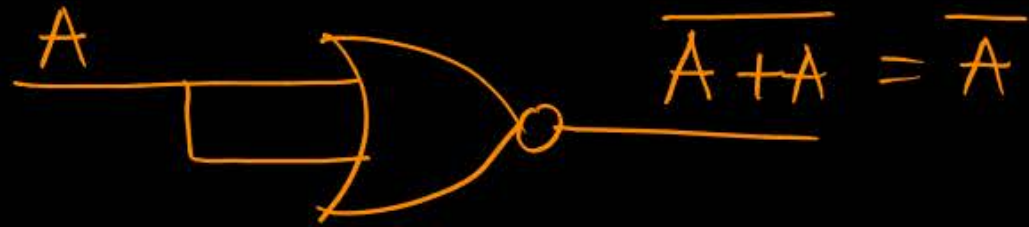
- IMP Points :

- If any one of the i/p line is at logic '1' then irrespective of other i/p lines o/p will be '0'.
- O/P will be '1' only if all the i/p lines together will be '0'.

[NOR GATE as Universal Circuit]

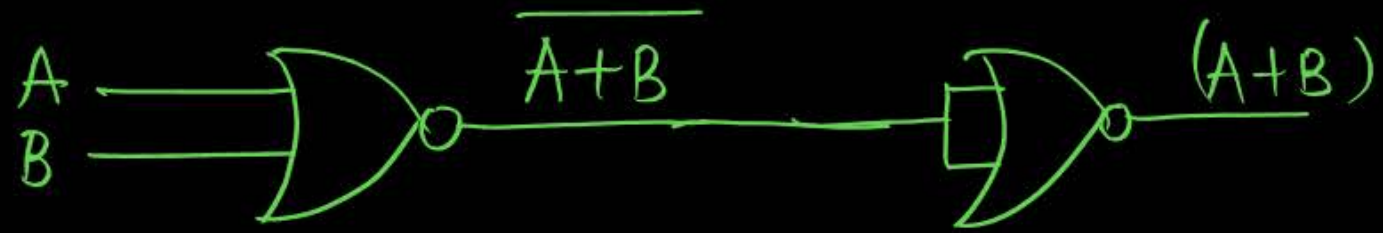


- NOT GATE

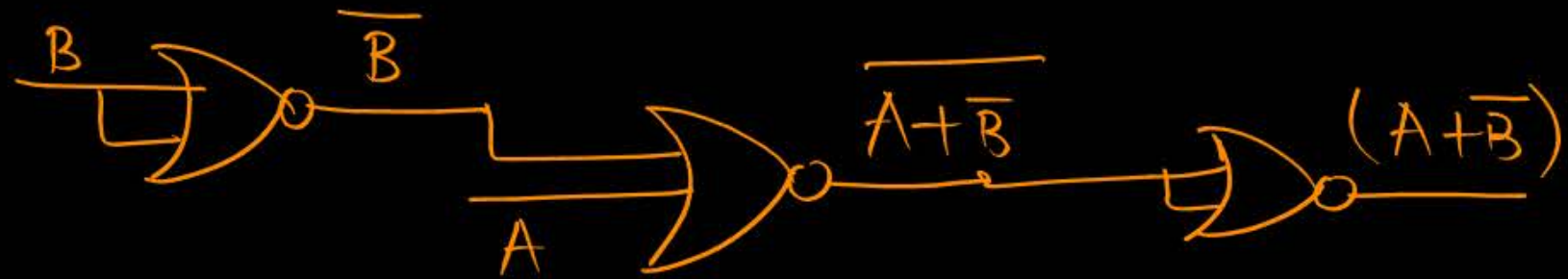


- OR GATE

$$A + B$$

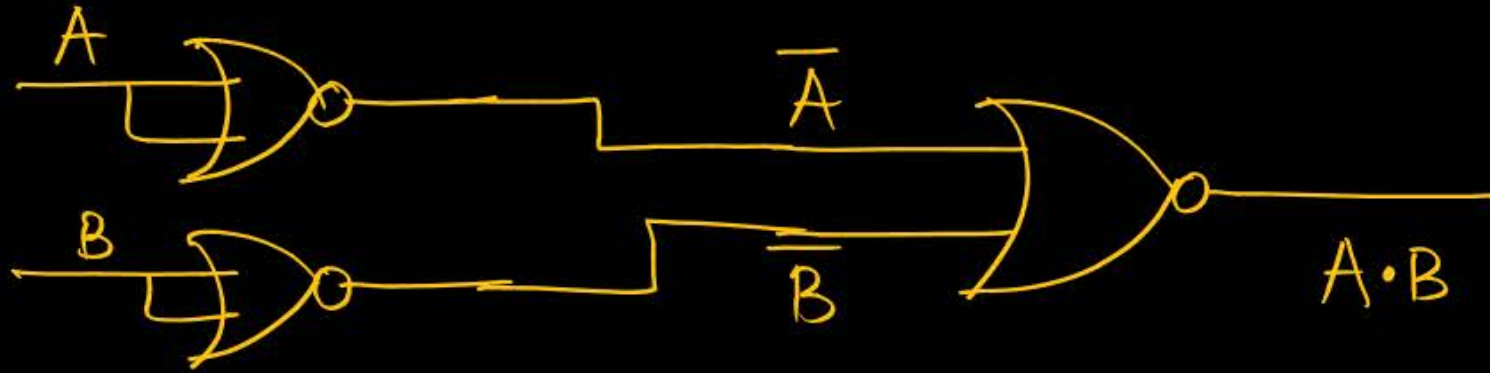


- $A + \overline{B}$

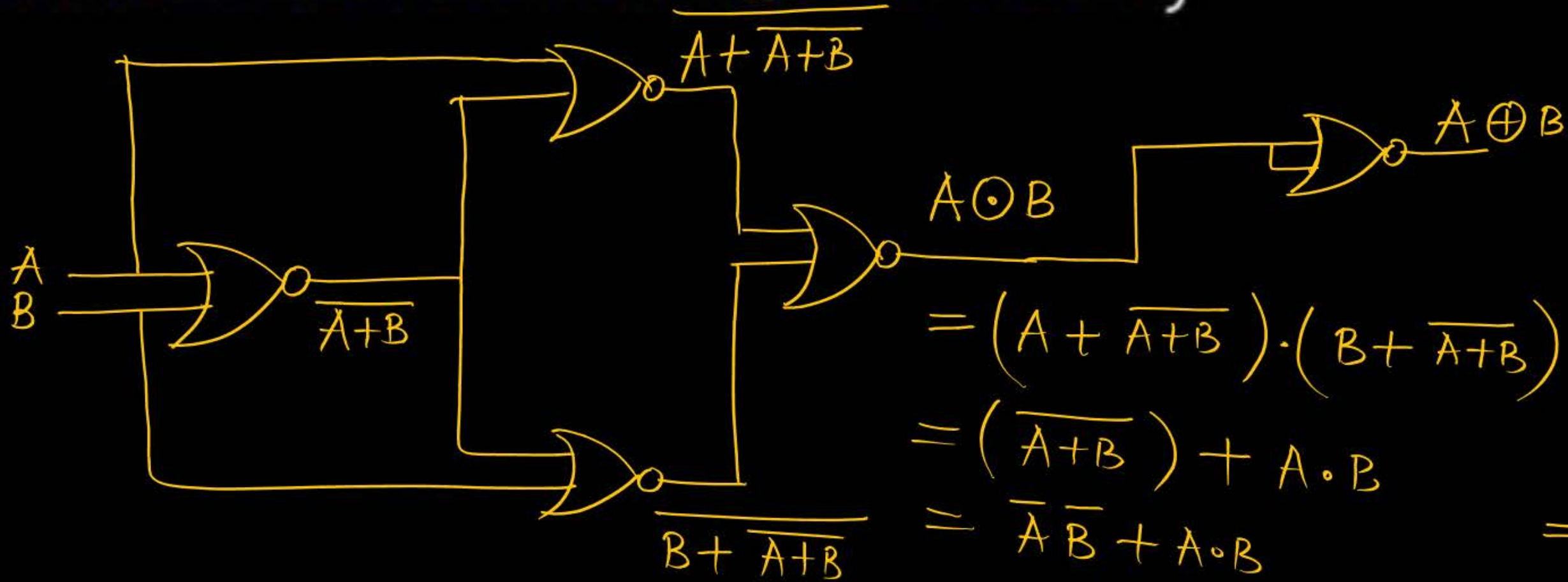


- AND GATE

$$A \cdot B$$



[XOR and XNOR GATE using NOR GATE]



$$\begin{aligned}
 &= (A + \overline{A+B}) \cdot (B + \overline{A+B}) = (A + \overline{A} \overline{B}) (B + \overline{A} \overline{B}) \\
 &= (\overline{A+B}) + A \cdot B \\
 &= \overline{A} \overline{B} + A \cdot B \\
 &= (A+B)(\overline{A}+\overline{B})
 \end{aligned}$$

GATES	No. of NAND	No. of NOR
NOT	1	1
AND	2	3
OR	3	2
XOR	4	5
XNOR	5	4

- $f(A, B, C) = A + BC \longrightarrow$ minimum no. of 2-I/P NOR requires to implement f 3.

$$f = (A + B) \cdot (A + C)$$

$$\overline{f} = \overline{A + B} + \overline{A + C}$$

$\downarrow$

$1(P)$

$\downarrow$

$1(Q)$

$$\overline{f} = P + Q$$

$$f = \overline{P + Q} \quad (1N)$$

$$\bullet f(A, B, C) = (A + \bar{B})(A + \bar{C}) = A + \underbrace{\bar{B} \cdot \bar{C}}_{\substack{\downarrow \\ 1N(P)}} = \underbrace{(A + P)}_{\substack{\downarrow \\ 2N}}$$

3

$$\bullet f(A, B, C) = (\bar{A} + B)(A + \bar{C}) = \underbrace{\bar{A} \bar{C}}_{\substack{\downarrow \\ 1N(R)}} + AB = R + A \cdot B = \underbrace{(R + A)(R + B)}_{\substack{\downarrow \\ 3N}}$$

4

$$\bullet f(A, B) = A \odot B = \bar{A} \bar{B} + AB = (\bar{A} + B)(A + \bar{B})$$

$$= \underbrace{R + AB}_{\substack{\downarrow \\ 1N}} = \underbrace{(R + A)(R + B)}_{\substack{\downarrow \\ 3N}}$$

- $f(A, B) = \overline{A}\overline{B} + \underline{A}B$

If compliments of variables ^{are} available then minimum no. of 2-i/p NOR gates requires to implement 3.

$$f(A, B) = (\overline{A} + B)(A + \overline{B})$$

$$= \underbrace{(P + B)(A + Q)}_{\downarrow 3}$$

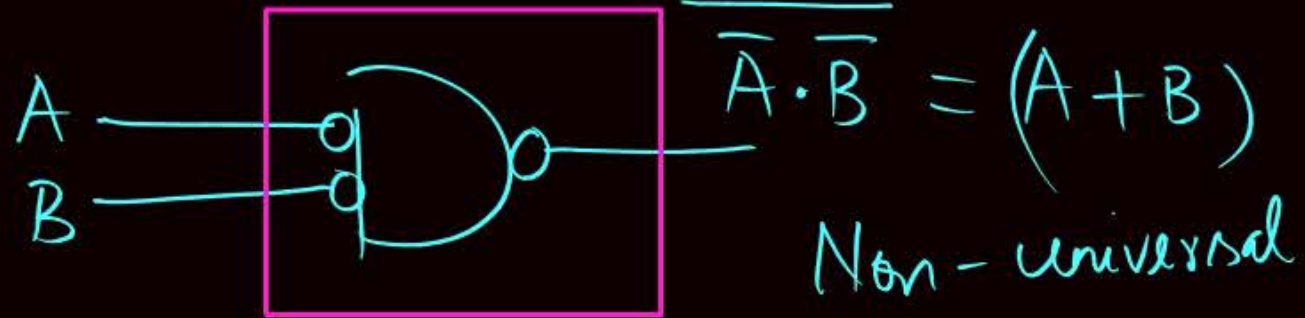
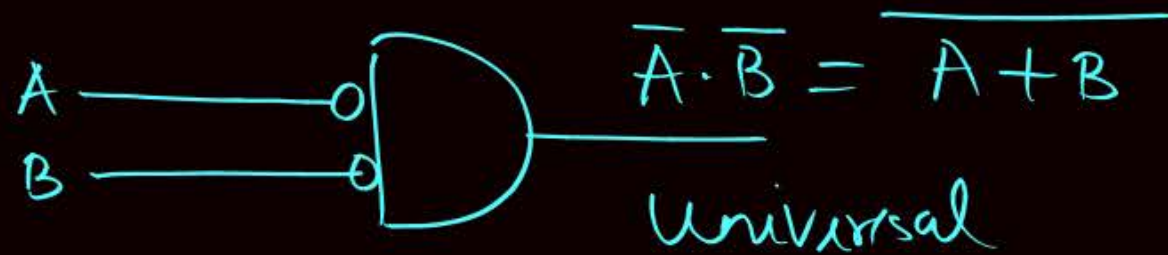
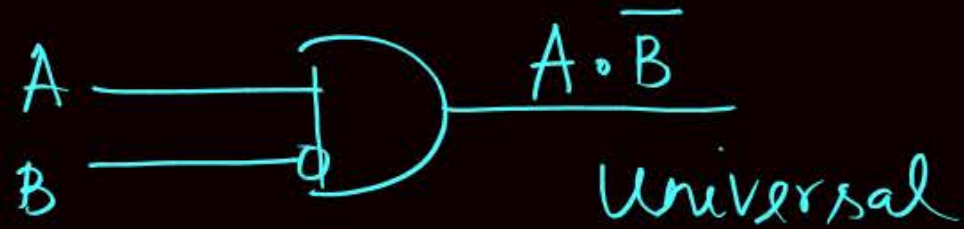
$$f(A, B, C, D) = \overline{A}\overline{B} + CD \quad \underline{4}$$

$$= \overbrace{R}^{\text{IN}} + CD$$

$$= \underbrace{(R + C)(R + D)}_{\downarrow 3}$$

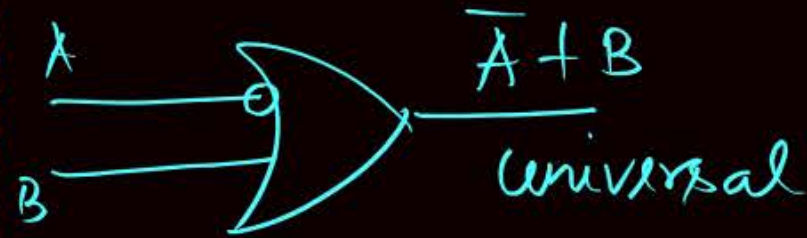
AND + NOT

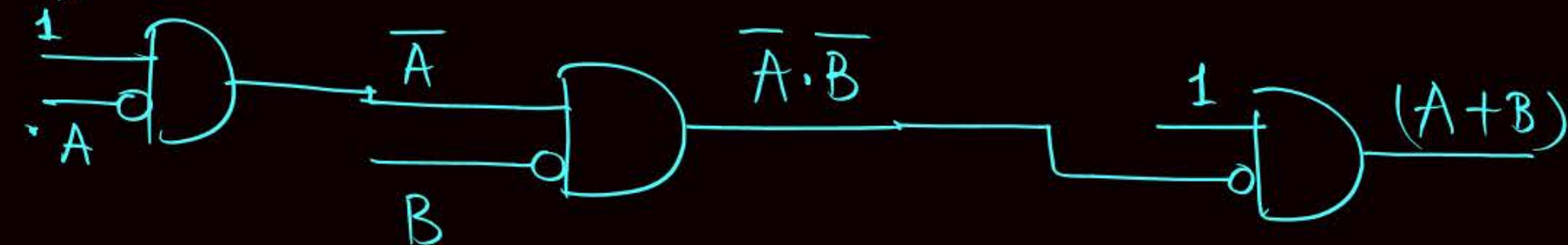
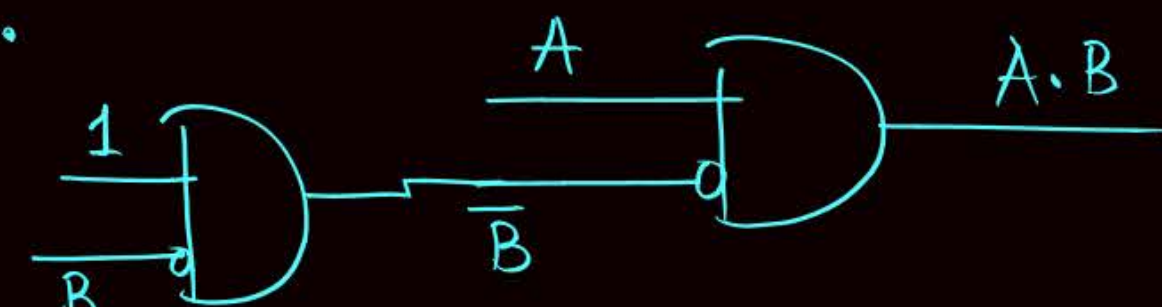
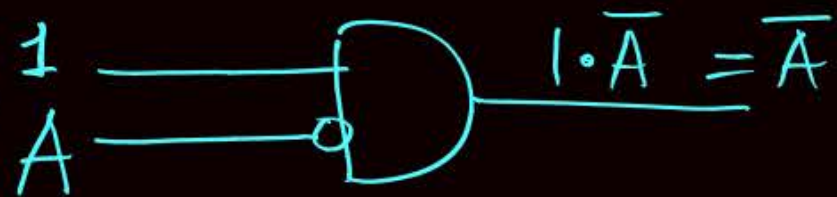
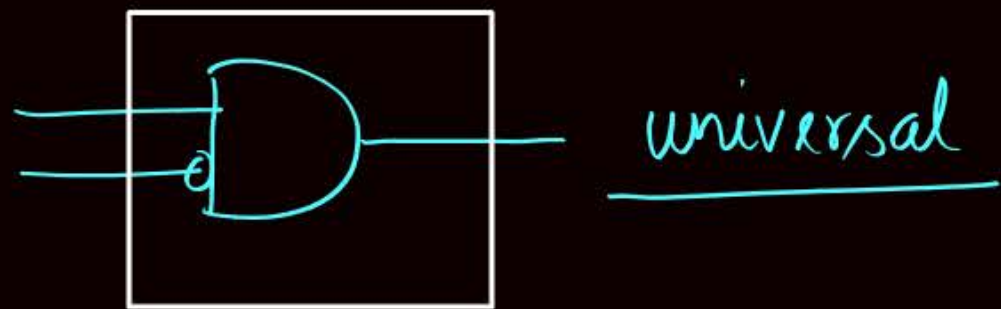
$$\overline{\overline{A} \cdot \overline{B}} = (A+B)$$



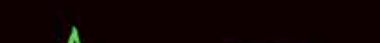
OR + NOT

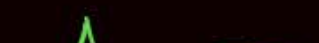
$$\overline{\overline{A} + \overline{B}} = A \cdot B$$





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•  $A \cdot \overline{0} = A$

•  $\overline{A} = 1$

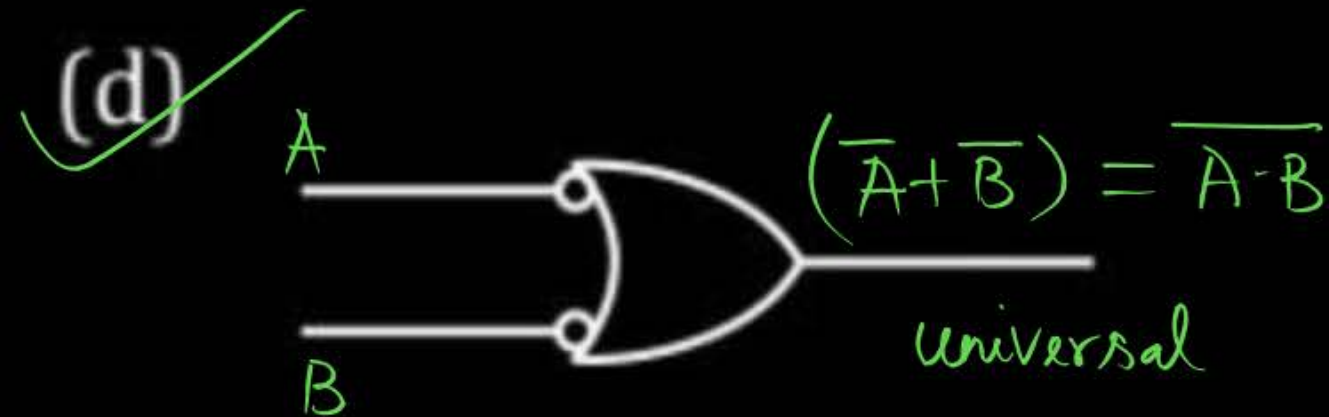
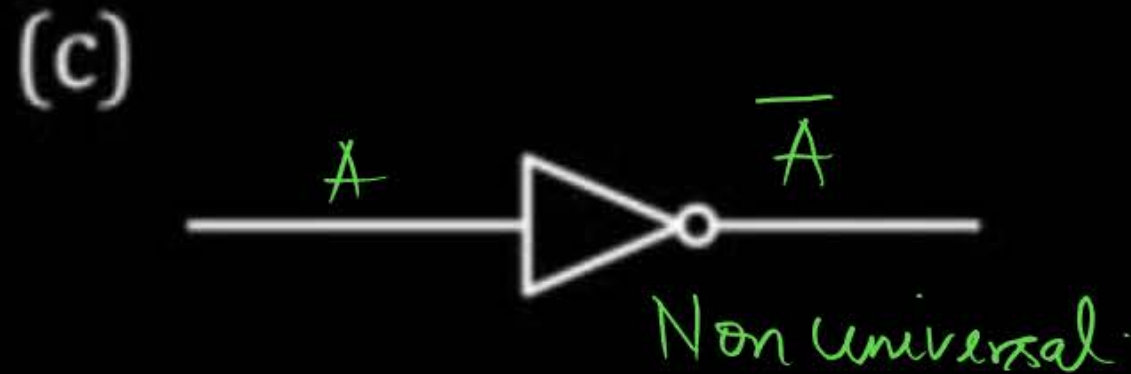
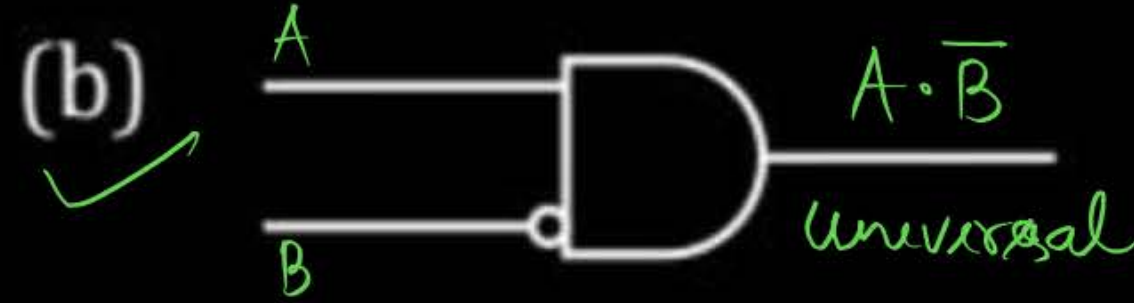
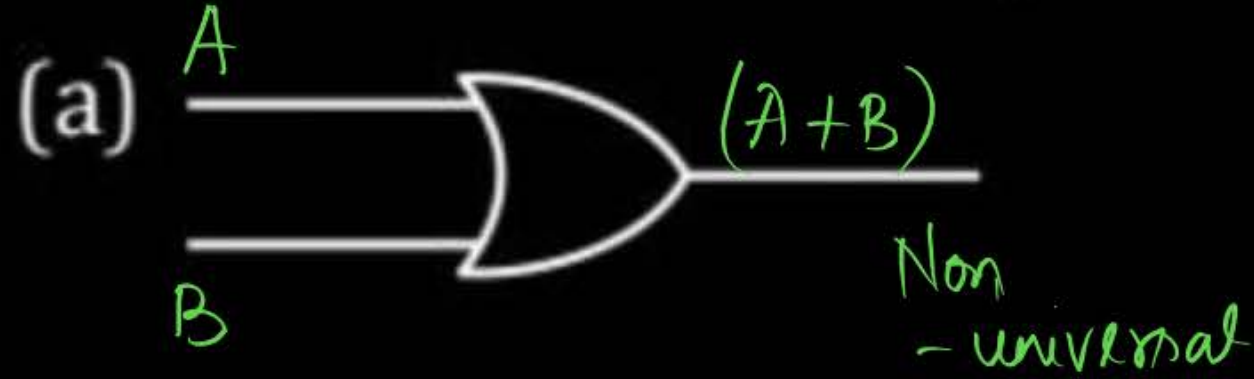
NOT operation is not possible

↓
Non-universal gate.

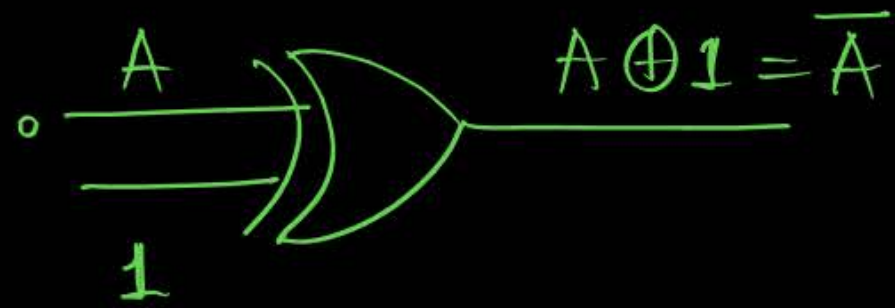
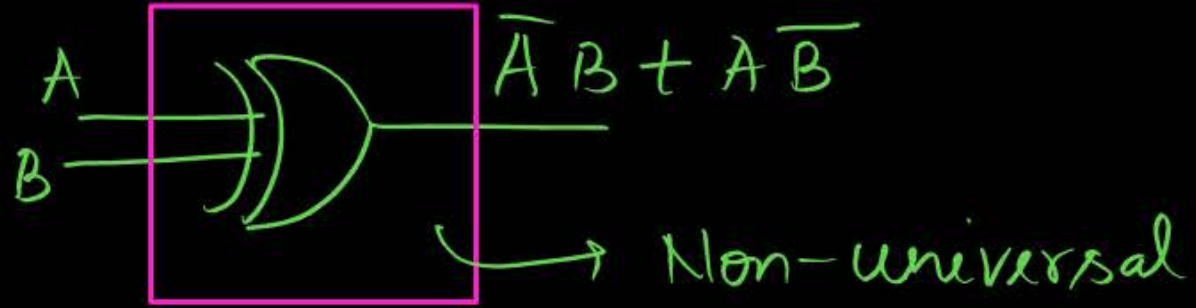
[**Question**] MSQ.



Which of the following circuit can work as universal gate ?



- $A \oplus B =$



$$A \oplus B \oplus \underline{\underline{AB}} = A + B$$

$$A \oplus B \oplus (A + B) = \bar{A} \cdot \bar{B}$$

- $(XOR, AND) \rightarrow \text{universal}$
- $(XOR, OR) \rightarrow \text{universal}$



2 Minute Summary



NAND & NOR.

Thank you

GW
Soldiers !

